Docker Volumes &
Data Persistence

Practical Assignment 8

Step 1:

```
C:\>docker --version
Docker version 29.6.1, build 8900f1d

C:\>docker info
Client:
 Version:    29.6.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.88.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.27.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.35.0-desktop.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.2.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
```

Step 2:

```
C:\>docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4f55086f7dd0: Pull complete
d5e71e642bf5: Download complete
Digest: sha256:7f4da0fc94bcece205a8c0b6f4d11c8196924654ffe5c4d1aa439b7f632048b2
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
 $ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
 https://hub.docker.com/

For more examples and ideas, visit:
 https://docs.docker.com/get-started/
```

Step 3:

```
C:\>docker run -dit --name data-test ubuntu
Unable to find image 'ubuntu:latest' locally
latest: Pulling from library/ubuntu
a7fb98a8eddd: Pull complete
617772c7d19b: Pull complete
cc2ffdbc1bf7: Download complete
Digest: sha256:678c6550cc43645e08669028bc177f50be4e7c5b8cca677067b1914d4afc7a03
Status: Downloaded newer image for ubuntu:latest
c35932c2aafd1ac93e5b76f42a720d3a139d92d56e4a25417c18394482f54ead

C:\>docker exec -it data-test bash
root@c35932c2aafd:/# mkdir /mydata
root@c35932c2aafd:/# echo "This is my important data" > /mydata/student.txt
root@c35932c2aafd:/# cat /mydata/student.txt
This is my important data
root@c35932c2aafd:/# exit
exit
```

Step 4:

```
C:\>docker rm -f data-test
data-test

C:\>docker run -dit --name data-test2 ubuntu
bd0235d29a1749b776d512bc0b2af7a009759c72765ffafa7cfc89a4a37001c5

C:\>docker exec -it data-test2 bash
root@bd0235d29a17:/# ls /mydata
ls: cannot access '/mydata': No such file or directory
root@bd0235d29a17:/# exit
exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container
    Learn more at https://docs.docker.com/go/debug-cli/

C:\>docker rm -f data-test2
data-test2

C:\>docker volume create student-data
student-data
```

Step 5:

```
C:\>docker volume ls
DRIVER    VOLUME NAME
local     5b482d62bee47b04d136567a76ba770b38a68cd1484d05b637ffaf8d13dc4b9a
local     5c7b5bbe569d608905c4caf13d8ff538c095849bde2f5b4db95d73b202ba2fde
local     7df0863600c568e37eaec8fc76a74df4d8ca8f5055df0b0ba2d95613959934a0
local     80cd0324d4294fcda91107796416202ea1ccc0e8380b689de0a06e104e3d97e9
local     255b5b5a6e7f61c16b3857b7b7c97fd931950d5bf1fc95ba31024737d8ebbaaf
local     6918fb0e2e3bf20aa03d79f07119b1673a7a5cc2503498b49c2ef1223f7842a7
local     58477e87c1034cc81c6f297d43fd992ada97479b94b169a5c98304f7834d5fb6
local     attendance_system_mongo_data
local     b59c40fd3415dd87892b0411bbb0e4433f8bc2068f9c4618bd7a921a165017f3
local     college-event-registration_mongo-data
local     jenkins_home
local     student-data

C:\>docker volume inspect student-data
[
    {
        "CreatedAt": "2026-08-08T05:10:42Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/student-data/_data",
        "Name": "student-data",
        "Options": null,
        "Scope": "local"
    }
]
```

Step 6:

```
C:\>docker run -dit --name volume-test -v student-data:/data ubuntu
d8a64a984183dc141da948dc0317ecdb14d3d2ceb7d2a58a1a78f373ba36ace1

C:\>docker exec -it volume-test bash
root@d8a64a984183:/# cd /data
root@d8a64a984183:/data# echo "Docker Volume Practical" > notes.txt
root@d8a64a984183:/data# echo "Student Name: Sharon" > student.txt
root@d8a64a984183:/data# ls
notes.txt  student.txt
root@d8a64a984183:/data# cat notes.txt
Docker Volume Practical
root@d8a64a984183:/data# cat student.txt
Student Name: Sharon
root@d8a64a984183:/data# exit
exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container
    Learn more at https://docs.docker.com/go/debug-cli/

C:\>docker rm -f volume-test
volume-test
```

Step 7:

```
C:\>docker volume ls
DRIVER    VOLUME NAME
local     5b482d62bee47b04d136567a76ba770b38a68cd1484d05b637ffaf8d13dc4b9a
local     5c7b5bbe569d608905c4caf13d8ff538c095849bde2f5b4db95d73b202ba2fde
local     7df0863600c568e37eaec8fc76a74df4d8ca8f5055df0b0ba2d95613959934a0
local     80cd0324d4294fcda91107796416202ea1ccc0e8380b689de0a06e104e3d97e9
local     255b5b5a6e7f61c16b3857b7b7c97fd931950d5bf1fc95ba31024737d8ebbaaf
local     6918fb0e2e3bf20aa03d79f07119b1673a7a5cc2503498b49c2ef1223f7842a7
local     58477e87c1034cc81c6f297d43fd992ada97479b94b169a5c98304f7834d5fb6
local     attendance_system_mongo_data
local     b59c40fd3415dd87892b0411bbb0e4433f8bc2068f9c4618bd7a921a165017f3
local     college-event-registration_mongo-data
local     jenkins_home
local     student-data

C:\>docker run -dit --name volume-test2 -v student-data:/data ubuntu
c1d135367ae107e923521482d2cda16e9bb3f7caed921650f4619a99122e9e3e

C:\>docker exec volume-test2 cat /data/notes.txt
Docker Volume Practical

C:\>docker exec volume-test2 cat /data/notes.txt
Docker Volume Practical

C:\>docker exec volume-test2 cat /data/student.txt
Student Name: Sharon

C:\>docker rm -f volume-test2
volume-test2

C:\>docker run -dit --name container1 -v student-data:/data ubuntu
a310207abadd905d88b7d631a9d23754ff136c3f491bfa651b2491cf9f5a5de4

C:\>docker run -dit --name container2 -v student-data:/data ubuntu
8f3c0b67026001db9abdab0705a7b67e45d4192f55f634865acea80703b9110e
```

Step 8:

```
C:\>docker exec container1 bash -c "echo 'Created by Container 1' > /data/container1.txt"

C:\>docker exec container2 cat /data/container1.txt
Created by Container 1

C:\>--mount source=student-data,target=/data
'--mount' is not recognized as an internal or external command,
operable program or batch file.

C:\>-v student-data:/data
'-v' is not recognized as an internal or external command,
operable program or batch file.

C:\>docker run -dit --name mount-test --mount source=student-data,target=/data ubuntu
3dd3c2d2d2bccaab9167e6c90bb9f3bab6a6d3c62f26d0bc36d233aa393c2981

C:\>docker exec mount-test ls /data
container1.txt
notes.txt
student.txt

C:\>docker run -dit --name readonly-test -v student-data:/data:ro ubuntu
f667ab81bfe015ce3aa40022dda5017cef92cec8cb4c1703dd82194d2892f667

C:\>docker exec -it readonly-test bash
root@f667ab81bfe0:/# cd /data
root@f667ab81bfe0:/data# echo "new data" > newfile.txt
bash: newfile.txt: Read-only file system
root@f667ab81bfe0:/data# exit
exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker de
    Learn more at https://docs.docker.com/go/debug-cli/

C:\>docker run -dit --name bind-test -v student-data:/data ubuntu
f01786d77eaef997735d3358d06947fca89dc59368e37343b7df5a36acf5eed0
```

Step 9:

```
C:\>docker run -dit --name bind-test -v C:\DockerLabs\data:/data ubuntu

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/bind-test" is already in use by container "f01786d77eaef997735d3358d06947fca8
9dc59368e37343b7df5a36acf5eed0". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

C:\>mkdir C:\DockerLabs\data
A subdirectory or file C:\DockerLabs\data already exists.

C:\>echo Hello from Windows > C:\DockerLabs\data\windows.txt

C:\>docker run -dit --name bind-test -v C:\DockerLabs\data:/data ubuntu

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/bind-test" is already in use by container "f01786d77eaef997735d3358d06947fca8
9dc59368e37343b7df5a36acf5eed0". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

C:\>docker exec -it bind-test bash
root@f01786d77eae:/# cat /data/windows.txt
cat: /data/windows.txt: No such file or directory
root@f01786d77eae:/# exit
exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug bind-test
    Learn more at https://docs.docker.com/go/debug-cli/
```

Step 10:

```
C:\>dir C:\DockerLabs\data
 Volume in drive C is Acer
 Volume Serial Number is 06AC-284D

 Directory of C:\DockerLabs\data

08-08-2026  11:00    <DIR>          .
08-08-2026  10:59    <DIR>          ..
08-08-2026  11:00                21 windows.txt
               1 File(s)             21 bytes
               2 Dir(s)  113,365,393,408 bytes free

C:\>type C:\DockerLabs\data\windows.txt
Hello from Windows

C:\>docker rm -f bind-test
bind-test

C:\>docker ps -a
CONTAINER ID   IMAGE                                  COMMAND                  CREATED          STATUS                      PORTS
 NAMES
f667ab81bfe0   ubuntu                                 "/bin/bash"              8 minutes ago    Up 8 minutes
 readonly-test
3dd3c2d2d2bc   ubuntu                                 "/bin/bash"              10 minutes ago   Up 10 minutes
 mount-test
8f3c0b670260   ubuntu                                 "/bin/bash"              12 minutes ago   Up 12 minutes
 container2
a310207abadd   ubuntu                                 "/bin/bash"              12 minutes ago   Up 12 minutes
 container1
17484ff58a54   hello-world                            "/hello"                 29 minutes ago   Exited (0) 29 minutes ago
 affectionate_blackburn
d1070b73c848   attendance_system-attendance-app       "python app.py"          3 days ago       Exited (0) 32 minutes ago
 attendance-app
e57a438df9a6   mongo:7                                "docker-entrypoint.s…"   3 days ago       Exited (0) 32 minutes ago
 mongodb
f1bef5c044af   jenkins/jenkins:lts                    "/usr/bin/tini -- /u…"   3 days ago       Exited (143) 2 days ago
 jenkins
81f8c5e8ebab   college-event-registration-app         "docker-entrypoint.s…"   3 weeks ago      Exited (255) 2 weeks ago    0.0.0.0:3000->3000/tcp
 college-event-registration-app-1
80245821f745   mongo:7                                "docker-entrypoint.s…"   4 weeks ago      Exited (0) 32 minutes ago
 college-event-registration-mongo-1
```

Step 11:

```
C:\>docker run -dit --name bind-test -v C:\DockerLabs\data:/data ubuntu
c590c366b0a0d831b0625dfc4c305c69cdd21793636d5a5c0cab2df7a628c18f

C:\>docker ps
CONTAINER ID   IMAGE     COMMAND       CREATED          STATUS          PORTS     NAMES
c590c366b0a0   ubuntu    "/bin/bash"   8 seconds ago    Up 7 seconds              bind-test
f667ab81bfe0   ubuntu    "/bin/bash"   8 minutes ago    Up 8 minutes              readonly-test
3dd3c2d2d2bc   ubuntu    "/bin/bash"   10 minutes ago   Up 10 minutes             mount-test
8f3c0b670260   ubuntu    "/bin/bash"   12 minutes ago   Up 12 minutes             container2
a310207abadd   ubuntu    "/bin/bash"   13 minutes ago   Up 13 minutes             container1

C:\>docker exec -it bind-test bash
root@c590c366b0a0:/# ls -l /data
total 0
-rwxrwxrwx 1 root root 21 Aug  8 05:30 windows.txt
root@c590c366b0a0:/# cat /data/windows.txt
Hello from Windows
root@c590c366b0a0:/# cat /data/windows.txt
Hello from Windows
root@c590c366b0a0:/# cat /data/windows.txt
Hello from Windows
root@c590c366b0a0:/# echo "Modified from Docker" > /data/docker.txt
root@c590c366b0a0:/# exit
exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug bind-test
    Learn more at https://docs.docker.com/go/debug-cli/

C:\>C:\DockerLabs\data
'C:\DockerLabs\data' is not recognized as an internal or external command,
operable program or batch file.

C:\>docker volume create website-data
website-data
```

Step 12:

```
C:\>docker run -d --name nginx-volume -p 8080:80 -v website-data:/usr/share/nginx/html nginx
Unable to find image 'nginx:latest' locally
latest: Pulling from library/nginx
5a4222b844e8: Pull complete
d84ae7b21412: Pull complete
f5de6e85ac74: Pull complete
c0df8d325117: Pull complete
3c55dc422a81: Pull complete
b8b80b9bc028: Pull complete
0f03cb4db0ef: Download complete
92fcf0fc2ef2: Download complete
Digest: sha256:8541484afbc9c8a5a8a99b379568ebbc957f658583ec9448fc43104229c03cf8
Status: Downloaded newer image for nginx:latest
4cbfeae40a7ef5f0df1e097a9982635b88b45e1d840f181a2cd9d16caaef4fa4

C:\>docker run --rm -v website-data:/data ubuntu bash -c "echo '<h1>Hello from Docker Volume</h1>' > /data/index.html"

C:\>docker rm -f nginx-volume
nginx-volume

C:\>docker volume ls
DRIVER    VOLUME NAME
local     5b482d62bee47b04d136567a76ba770b38a68cd1484d05b637ffaf8d13dc4b9a
local     5c7b5bbe569d608905c4caf13d8ff538c095849bde2f5b4db95d73b202ba2fde
local     7df0863600c568e37eaec8fc76a74df4d8ca8f5055df0b0ba2d95613959934a0
local     80cd0324d4294fcda91107796416202ea1ccc0e8380b689de0a06e104e3d97e9
local     255b5b5a6e7f61c16b3857b7b7c97fd931950d5bf1fc95ba31024737d8ebbaaf
local     6918fb0e2e3bf20aa03d79f07119b1673a7a5cc2503498b49c2ef1223f7842a7
local     58477e87c1034cc81c6f297d43fd992ada97479b94b169a5c98304f7834d5fb6
local     attendance_system_mongo_data
local     b59c40fd3415dd87892b0411bbb0e4433f8bc2068f9c4618bd7a921a165017f3
local     college-event-registration_mongo-data
local     jenkins_home
local     student-data
local     website-data

C:\>docker run -d --name nginx-new -p 8080:80 -v website-data:/usr/share/nginx/html nginx
59d3f9c4ca6f60a8c0c70a1ac891002ccf8143295b94d7034005ccefde191b9b

C:\>docker volume create mongo-data
mongo-data
```

Step 13:

```
C:\>docker run -d --name mongodb -p 27017:27017 -v mongo-data:/data/db mongo

What's next:
    Debug this container error with Gordon → docker ai "help me fix this container error"
docker: Error response from daemon: Conflict. The container name "/mongodb" is already in use by container "e57a438df9a6ac4432e7ea42a5ba5e9df682
ab796c9a1a15e83ffbabee4fd1a2". You have to remove (or rename) that container to be able to reuse that name.

Run 'docker run --help' for more information

C:\>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS              PORTS                                    NAMES
59d3f9c4ca6f   nginx     "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp   nginx-new
c590c366b0a0   ubuntu    "/bin/bash"              6 minutes ago        Up 6 minutes                                                 bind-test
f667ab81bfe0   ubuntu    "/bin/bash"              15 minutes ago       Up 15 minutes                                                readonly-te
st
3dd3c2d2d2bc   ubuntu    "/bin/bash"              17 minutes ago       Up 17 minutes                                                mount-test
8f3c0b670260   ubuntu    "/bin/bash"              19 minutes ago       Up 19 minutes                                                container2
a310207abadd   ubuntu    "/bin/bash"              19 minutes ago       Up 19 minutes                                                container1

C:\>docker start mongodb
mongodb

C:\>docker ps
CONTAINER ID   IMAGE     COMMAND                  CREATED              STATUS              PORTS                                          NAM
ES
59d3f9c4ca6f   nginx     "/docker-entrypoint.…"   About a minute ago   Up About a minute   0.0.0.0:8080->80/tcp, [::]:8080->80/tcp         ngi
nx-new
c590c366b0a0   ubuntu    "/bin/bash"              6 minutes ago        Up 6 minutes                                                       bin
d-test
f667ab81bfe0   ubuntu    "/bin/bash"              15 minutes ago       Up 15 minutes                                                      rea
donly-test
3dd3c2d2d2bc   ubuntu    "/bin/bash"              17 minutes ago       Up 17 minutes                                                      mou
nt-test
8f3c0b670260   ubuntu    "/bin/bash"              19 minutes ago       Up 19 minutes                                                      con
tainer2
a310207abadd   ubuntu    "/bin/bash"              19 minutes ago       Up 19 minutes                                                      con
tainer1
e57a438df9a6   mongo:7   "docker-entrypoint.s…"   3 days ago           Up 3 seconds        0.0.0.0:27017->27017/tcp, [::]:27017->27017/tcp   mon
godb

C:\>docker rm -f mongodb
mongodb
```

Step 14:

```
C:\>docker run -d --name mongodb-new -p 27017:27017 -v mongo-data:/data/db mongo
e9f21f133fe1dd81fd939f6d3bbcae7c2391e3c3ba4c9f094a63364c92b5e8a8

C:\>mkdir C:\DockerLabs\backup

C:\>docker run --rm -v student-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar czf /backup/student-data.tar.gz -C /data .

C:\>docker volume create restored-data
restored-data

C:\>docker run --rm -v restored-data:/data -v C:\DockerLabs\backup\:/backup ubuntu tar xzf /backup/student-data.tar.gz -C /data

C:\>docker run --rm -v restored-data:/data ubuntu ls -l /data
total 12
-rw-r--r-- 1 root root 23 Aug  8 05:21 container1.txt
-rw-r--r-- 1 root root 24 Aug  8 05:13 notes.txt
-rw-r--r-- 1 root root 21 Aug  8 05:13 student.txt

C:\>docker volume ls
DRIVER    VOLUME NAME
local     5b482d62bee47b04d136567a76ba770b38a68cd1484d05b637ffaf8d13dc4b9a
local     5c7b5bbe569d608905c4caf13d8ff538c095849bde2f5b4db95d73b202ba2fde
local     7df0863600c568e37eaec8fc76a74df4d8ca8f5055df0b0ba2d95613959934a0
local     80cd0324d4294fcda91107796416202ea1ccc0e8380b689de0a06e104e3d97e9
local     255b5b5a6e7f61c16b3857b7b7c97fd931950d5bf1fc95ba31024737d8ebbaaf
local     6918fb0e2e3bf20aa03d79f07119b1673a7a5cc2503498b49c2ef1223f7842a7
local     58477e87c1034cc81c6f297d43fd992ada97479b94b169a5c98304f7834d5fb6
local     a12cdc16b3e8ef5c721ff0e898f6de810e588a919e9196d31d7ac1177ba405a2
local     attendance_system_mongo_data
local     b59c40fd3415dd87892b0411bbb0e4433f8bc2068f9c4618bd7a921a165017f3
local     college-event-registration_mongo-data
local     jenkins_home
local     mongo-data
local     restored-data
local     student-data
local     website-data

C:\>docker volume create myvolume
myvolume
```

Step 15:

```
C:\>docker volume inspect myvolume
[
    {
        "CreatedAt": "2026-08-08T05:50:14Z",
        "Driver": "local",
        "Labels": null,
        "Mountpoint": "/var/lib/docker/volumes/myvolume/_data",
        "Name": "myvolume",
        "Options": null,
        "Scope": "local"
    }
]

C:\>docker volume rm myvolume
myvolume

C:\>docker volume prune
WARNING! This will remove anonymous local volumes not used by at least one container.
Are you sure you want to continue? [y/N] y
Deleted Volumes:
58477e87c1034cc81c6f297d43fd992ada97479b94b169a5c98304f7834d5fb6
5c7b5bbe569d608905c4caf13d8ff538c095849bde2f5b4db95d73b202ba2fde
255b5b5a6e7f61c16b3857b7b7c97fd931950d5bf1fc95ba31024737d8ebbaaf
5b482d62bee47b04d136567a76ba770b38a68cd1484d05b637ffaf8d13dc4b9a
6918fb0e2e3bf20aa03d79f07119b1673a7a5cc2503498b49c2ef1223f7842a7
80cd0324d4294fcda91107796416202ea1ccc0e8380b689de0a06e104e3d97e9
b59c40fd3415dd87892b0411bbb0e4433f8bc2068f9c4618bd7a921a165017f3

Total reclaimed space: 210.3MB

C:\>docker volume create student-data
student-data

C:\>docker run -v student-data:/data ubuntu
```

Step 16:

```
C:\>docker exec container2 ls /data
container1.txt
notes.txt
student.txt

C:\>docker volume create student-records
student-records

C:\>docker run -dit --name student-app -v student-records:/data ubuntu
2b66ff307c13b3a77fc20afda126a9b710ba91000f290a8f391f6647ce1ff415

C:\>docker exec -it student-app bash
root@2b66ff307c13:/# echo "101, Rahul, BCA" > /data/students.csv
root@2b66ff307c13:/# echo "102, Priya, BCA" >> /data/students.csv
root@2b66ff307c13:/# cat /data/students.csv
101, Rahul, BCA
102, Priya, BCA
root@2b66ff307c13:/# exit
exit

What's next:
    Try Docker Debug for seamless, persistent debugging tools in any container or image → docker debug student-app
    Learn more at https://docs.docker.com/go/debug-cli/

C:\>docker rm -f student-app
student-app

C:\>docker run -dit --name student-app-new -v student-records:/data ubuntu
f0709a43113000eeb8e56dc988159384782de330ae1b401eb28c54797bbfd855

C:\>docker exec student-app-new cat /data/students.csv
101, Rahul, BCA
102, Priya, BCA
```

Step 17:

```
C:\>docker info
Client:
 Version:    29.6.1
 Context:    desktop-linux
 Debug Mode: false
 Plugins:
  agent: Docker AI Agent Runner (Docker Inc.)
    Version:  v1.88.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-agent.exe
  ai: Docker AI Agent - Ask Gordon (Docker Inc.)
    Version:  v1.27.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-ai.exe
  buildx: Docker Buildx (Docker Inc.)
    Version:  v0.35.0-desktop.2
    Path:     C:\Program Files\Docker\cli-plugins\docker-buildx.exe
  compose: Docker Compose (Docker Inc.)
    Version:  v5.2.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-compose.exe
  debug: Get a shell into any image or container (Docker Inc.)
    Version:  0.0.47
    Path:     C:\Program Files\Docker\cli-plugins\docker-debug.exe
  desktop: Docker Desktop commands (Docker Inc.)
    Version:  v0.4.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-desktop.exe
  dhi: CLI for managing Docker Hardened Images (Docker Inc.)
    Version:  v0.0.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-dhi.exe
  extension: Manages Docker extensions (Docker Inc.)
    Version:  v0.2.31
    Path:     C:\Program Files\Docker\cli-plugins\docker-extension.exe
  init: Creates Docker-related starter files for your project (Docker Inc.)
    Version:  v1.4.0
    Path:     C:\Program Files\Docker\cli-plugins\docker-init.exe
  mcp: Docker MCP Plugin (Docker Inc.)
    Version:  v0.43.1
    Path:     C:\Program Files\Docker\cli-plugins\docker-mcp.exe
  model: Docker Model Runner (Docker Inc.)
    Version:  v1.2.5
    Path:     C:\Program Files\Docker\cli-plugins\docker-model.exe
  offload: Docker Offload (Docker Inc.)
    Version:  v0.6.7
    Path:     C:\Program Files\Docker\cli-plugins\docker-offload.exe
```

Step 18:

```
Cgroup Driver: cgroupfs
Cgroup Version: 2
Plugins:
 Volume: local
 Network: bridge host ipvlan macvlan null overlay
 Log: awslogs fluentd gcplogs gelf journald json-file local splunk syslog
CDI spec directories:
 /etc/cdi
 /var/run/cdi
Discovered Devices:
 cdi: docker.com/gpu=webgpu
Swarm: inactive
Runtimes: io.containerd.runc.v2 nvidia runc
Default Runtime: runc
Init Binary: docker-init
containerd version: e53c7c1516c3b2bff98eb76f1f4117477e6f4e66
runc version: v1.3.6-0-g491b69ba
init version: de40ad0
Security Options:
 seccomp
  Profile: builtin
 cgroupns
Kernel Version: 6.6.87.2-microsoft-standard-WSL2
Operating System: Docker Desktop
OSType: linux
Architecture: x86_64
CPUs: 24
Total Memory: 7.61GiB
Name: docker-desktop
ID: 7be5c3ac-d288-4e72-8439-dff2754d1fd2
Docker Root Dir: /var/lib/docker
Debug Mode: false
HTTP Proxy: http.docker.internal:3128
HTTPS Proxy: http.docker.internal:3128
No Proxy: hubproxy.docker.internal
Labels:
 com.docker.desktop.address=npipe://\\.\pipe\docker_cli
Experimental: false
Insecure Registries:
 hubproxy.docker.internal:5555
 ::1/128
 127.0.0.0/8
Live Restore Enabled: false
Firewall Backend: iptables
```